

Jesse Diaz Thaler

Publications and Preprints

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† = Paper arising from a supervised B.S. or M.Eng. thesis

§ = Authors ordered by contribution

- [147] * Gregorio de la Fuente and Jesse Thaler, *Simplex Demixing: Disentangling Multiple Light-Flavor Jets at Colliders* [arXiv:2607.24921].
- [146] Benjamin Nachman and Jesse Thaler, *The Well-Tempered Likelihood: Honest Confidence Intervals for Misspecified Models* [arXiv:2607.20718].
- [145] § Rikab Gambhir, Luisa Lucie-Smith, and Jesse Thaler, *Interpreting 'Interpretability' and Explaining 'Explainability' in Machine Learning in Physics* [arXiv:2606.26228].
- [144] § Umar Sohail Qureshi, Krish Desai, Jesse Thaler, and Benjamin Nachman, *Reweighting Adversarial Networks for Unbinned Unfolding* [arXiv:2606.06603].
- [143] Ezequiel Alvarez, Sean Benevedes, Manuel Szwec, and Jesse Thaler, *Many Wrongs Make a Right: Leveraging Biased Simulations Towards Unbiased Parameter Inference* [arXiv:2604.02219].
- [142] Samuel Bright-Thonney, Thomas R. Harvey, Andre Lukas, and Jesse Thaler, *Sven: Singular Value Descent as a Computationally Efficient Natural Gradient Method* [arXiv:2604.01279].
- [141] Nathan Benjamin, A. Liam Fitzpatrick, Wei Li, and Jesse Thaler, *Descending into the Modular Bootstrap* [arXiv:2604.01275].
- [140] Benoît Assi, Kyle Lee, and Jesse Thaler, *Improving Parton Shower Predictions via Precision Moments of Energy Flow Polynomials* [arXiv:2604.00084].
- [139] Andrew Ferguson, Marisa LaFleur, Lars Ruthotto, Jesse Thaler, Yuan-Sen Ting, Pratyush Tiwary, Soledad Villar, et al., *The Future of Artificial Intelligence and the Mathematical and Physical Sciences (AI+MPS)*, MLST 7:023001 (2026) [arXiv:2509.02661].
- [138] * Sean Benevedes and Jesse Thaler, *Frequentist Uncertainties on Neural Density Ratios with WiFi Ensembles*, Phys. Rev. D112:056024 (2025) [arXiv:2506.00113].
- [137] * Rikab Gambhir, Radha Mastandrea, Benjamin Nachman, Jesse Thaler, *Isolating Unisolated Upsilons with Anomaly Detection in CMS Open Data*, Phys. Rev. Lett. 135:021902 (2025) [arXiv:2502.14036].
- [136] Benoît Assi, Stefan Höche, Kyle Lee, and Jesse Thaler, *QCD Theory Meets Information Theory*, Phys. Rev. Lett. 135:131901 (2025) [arXiv:2501.17219].

- [135] Johann Brehmer, Víctor Bresó, Pim de Haan, Tilman Plehn, Huilin Qu, Jonas Spinner, and Jesse Thaler, *A Lorentz-Equivariant Transformer for All of the LHC*, SciPost Physics 19:108 (2025) [arXiv:2411.00446].
- [134] Jesse Thaler and Sokratis Trifinopoulos, *Flavor Patterns of Fundamental Particles from Quantum Entanglement?*, Phys. Rev. D111:056021 (2025) [arXiv:2410.23343].
- [133] * Samuel Alipour-fard, Ankita Budhraj, Jesse Thaler, and Wouter J. Waalewijn, *New Angles on Energy Correlators*, Phys. Rev. Lett. 134:231902 (2025) [arXiv:2410.16368].
- [132] * Rikab Gambhir, Andrew J. Larkoski, and Jesse Thaler, *SPECTER: Efficient Evaluation of the Spectral EMD*, JHEP 2412:219 (2024) [arXiv:2410.05379].
- [131] Krish Desai, Benjamin Nachman, and Jesse Thaler, *Moment Unfolding*, Phys. Rev. D110:116013 (2024) [arXiv:2407.11284].
- [130] § Jonas Spinner, Victor Bresó, Pim de Haan, Tilman Plehn, Jesse Thaler, and Johann Brehmer, *Lorentz-Equivariant Geometric Algebra Transformers for High-Energy Physics*, NeurIPS 2024 [arXiv:2405.14806].
- [129] * Rikab Gambhir, Athis Osathapan, and Jesse Thaler, *Moments of Clarity: Streamlining Latent Spaces in Machine Learning using Moment Pooling*, Phys. Rev. D110:074020 (2024) [arXiv:2403.08854].
- [128] § Siddharth Mishra-Sharma, Yiding Song, and Jesse Thaler, *PAPERCLIP: Associating Astronomical Observations and Natural Language with Multi-Modal Models*, Conf. Lang. Mod. 2024 [arXiv:2403.08851].
- [127] † Eric M. Metodiev, Jesse Thaler, and Raymond Wynne, *Anomaly Detection in Collider Physics via Factorized Observables*, Phys. Rev. D110:055012 (2024) [arXiv:2312.00119].
- [126] Samuel Bright-Thonney, Benjamin Nachman, and Jesse Thaler, *Safe but Incalculable: Energy-Weighting is Not All You Need*, Phys. Rev. D110:014029 (2024) [arXiv:2311.07652].
- [125] Fabrizio Caola, Radosław Grabarczyk, Maxwell L. Hutt, Gavin P. Salam, Ludovic Scyboz, and Jesse Thaler, *Flavoured Jets with Exact Anti-kt Kinematics and Tests of Infrared and Collinear Safety*, Phys. Rev. D108:094010 (2023) [arXiv:2306.07314].
- [124] Andrew J. Larkoski and Jesse Thaler, *A Spectral Metric for Collider Geometry*, JHEP 2308:107 (2023) [arXiv:2305.03751].
- [123] * Samuel Alipour-fard, Patrick T. Komiske, Eric M. Metodiev, and Jesse Thaler, *Pileup and Infrared Radiation Annihilation (PIRANHA): A Paradigm for Continuous Jet Grooming*, JHEP 2309:157 (2023) [arXiv:2305.00989].
- [122] * Demba Ba, Akshunna S. Dogra, Rikab Gambhir, Abiy Tasissa, and Jesse Thaler, *SHAPER: Can You Hear the Shape of a Jet?*, JHEP 2306:195 (2023) [arXiv:2302.12266].
- [121] Erik Buhmann, Gregor Kasieczka, and Jesse Thaler, *EPiC-GAN: Equivariant Point Cloud Generation for Particle Jets*, SciPost Phys. 15:130 (2023) [arXiv:2301.08128].
- [120] Peter Onyisi, Delon Shen, and Jesse Thaler, *Comparing Point Cloud Strategies for Collider Event Classification*, Phys. Rev. D108:012001 (2023) [arXiv:2212.10659].

- [119] [†] Eric R. Anschuetz, Lena Funcke, Patrick T. Komiske, Serhii Kryhin, and Jesse Thaler, *Degeneracy Engineering for Classical and Quantum Annealing: A Case Study of Sparse Linear Regression in Collider Physics*, Phys. Rev. D106:056008 (2022) [arXiv:2205.10375].
- [118] Pedro Cal, Jesse Thaler, and Wouter J. Waalewijn, *Power Counting Energy Flow Polynomials*, JHEP 2209:021 (2022) [arXiv:2205.06818].
- [117] * Rikab Gambhir, Benjamin Nachman, and Jesse Thaler, *Bias and Priors in Machine Learning Calibrations for High Energy Physics*, Phys. Rev. D106:036011 (2022) [arXiv:2205.05084].
- [116] [†] Patrick T. Komiske, Serhii Kryhin, and Jesse Thaler, *Disentangling Quarks and Gluons with CMS Open Data*, Phys. Rev. D106:094021 (2022) [arXiv:2205.04459].
- [115] * Rikab Gambhir, Benjamin Nachman, and Jesse Thaler, *Learning Uncertainties the Frequentist Way: Calibration and Correlation in High Energy Physics*, Phys. Rev. Lett. 129:082001 (2022) [arXiv:2205.03413].
- [114] Hao Chen, Ian Moutt, Jesse Thaler, and Hua Xing Zhu, *Non-Gaussianities in Collider Energy Flux*, JHEP 2207:146 (2022) [arXiv:2205.02857].
- [113] Andrea Delgado and Jesse Thaler, *Quantum Annealing for Jet Clustering with Thrust*, Phys. Rev. D106:094016 (2022) [arXiv:2205.02814].
- [112] Patrick T. Komiske, Ian Moutt, Jesse Thaler, and Hua Xing Zhu, *Analyzing N-point Energy Correlators Inside Jets with CMS Open Data*, Phys. Rev. Lett. 130:051901 (2023) [arXiv:2201.07800].
- [111] Krish Desai, Benjamin Nachman, and Jesse Thaler, *SymmetryGAN: Symmetry Discovery with Deep Learning*, Phys. Rev. D105:096031 (2022) [arXiv:2112.05722].
- [110] Benjamin Nachman and Jesse Thaler, *Neural Conditional Reweighting*, Phys. Rev. D105:076015 (2022) [arXiv:2107.08979].
- [109] Benjamin Nachman and Jesse Thaler, *E Pluribus Unum Ex Machina: Learning from Many Collider Events at Once*, Phys. Rev. D103:116013 (2021) [arXiv:2101.07263].
- [108] Taylor Faucett, Jesse Thaler, and Daniel Whiteson, *Mapping Machine-Learned Physics into a Human-Readable Space*, Phys. Rev. D103:036020 (2021) [arXiv:2010.11998].
- [107] Jasmine Brewer, Jesse Thaler, and Andrew P. Turner, *Data-Driven Quark and Gluon Jet Modification in Heavy-Ion Collisions*, Phys. Rev. C103:L021901 (2021) [arXiv:2008.08596].
- [106] Benjamin Nachman and Jesse Thaler, *Neural Resampler for Monte Carlo Reweighting with Preserved Uncertainties*, Phys. Rev. D102:076004 (2020) [arXiv:2007.11586].
- [105] Cari Cesarotti and Jesse Thaler, *A Robust Measure of Event Isotropy at Colliders*, JHEP 2008:084 (2020) [arXiv:2004.06125].
- [104] * Patrick T. Komiske, Eric M. Metodiev, and Jesse Thaler, *The Hidden Geometry of Particle Collisions*, JHEP 2007:006 (2020) [arXiv:2004.04159].
- [103] * Anders Andreassen, Patrick T. Komiske, Eric M. Metodiev, Benjamin Nachman, and Jesse Thaler, *OmniFold: A Method to Simultaneously Unfold All Observables*, Phys. Rev. Lett. 124:182001 (2020) [arXiv:1911.09107].

- [102] * Patrick T. Komiske, Eric M. Metodiev, and Jesse Thaler, *Cutting Multiparticle Correlators Down to Size*, Phys. Rev. D101:036019 (2020) [arXiv:1911.04491].
- [101] Timothy Cohen, Gilly Elor, Andrew J. Larkoski, and Jesse Thaler, *Circumnavigating Collinear Superspace*, JHEP 2002:156 (2020) [arXiv:1909.00009].
- [100] § Annie Y. Wei, Preksha Naik, Aram W. Harrow, and Jesse Thaler, *Quantum Algorithms for Jet Clustering*, Phys. Rev. D101:094015 (2020) [arXiv:1908.08949].
- [99] *† Patrick T. Komiske, Radha Mastandrea, Eric M. Metodiev, Preksha Naik, and Jesse Thaler, *Exploring the Space of Jets with CMS Open Data*, Phys. Rev. D101:034009 (2020) [arXiv:1908.08542].
- [98] Cari Cesarotti, Yotam Soreq, Matthew J. Strassler, Jesse Thaler, and Wei Xue, *Searching in CMS Open Data for Dimuon Resonances with Substantial Transverse Momentum*, Phys. Rev. D100:015021 (2019) [arXiv:1902.04222].
- [97] * Patrick T. Komiske, Eric M. Metodiev, and Jesse Thaler, *The Metric Space of Collider Events*, Phys. Rev. Lett. 123:041801 (2019) [arXiv:1902.02346].
- [96] Jasmine Brewer, José Guilherme Milhano, and Jesse Thaler, *Sorting Out Quenched Jets*, Phys. Rev. Lett. 122:222301 (2019) [arXiv:1812.05111].
- [95] Timothy Cohen, Gilly Elor, Andrew J. Larkoski, and Jesse Thaler, *Navigating Collinear Superspace*, JHEP 2002:146 (2020) [arXiv:1810.11032].
- [94] * Patrick T. Komiske, Eric M. Metodiev, and Jesse Thaler, *Energy Flow Networks: Deep Sets for Particle Jets*, JHEP 1901:121 (2019) [arXiv:1810.05165].
- [93] § Hongwan Liu, Brodi D. Elwood, Matthew Evans, and Jesse Thaler, *Searching for Axion Dark Matter with Birefringent Cavities*, Phys. Rev. D100:023548 (2019) [arXiv:1809.01656].
- [92] * Patrick T. Komiske, Eric M. Metodiev, and Jesse Thaler, *An Operational Definition of Quark and Gluon Jets*, JHEP 1811:059 (2018) [arXiv:1809.01140].
- [91] † Eleanor Hall and Jesse Thaler, *Photon Isolation and Jet Substructure*, JHEP 1809:164 (2018) [arXiv:1805.11622].
- [90] * Benjamin T. Elder and Jesse Thaler, *Aspects of Track-Assisted Mass*, JHEP 1903:104 (2019) [arXiv:1805.11109].
- [89] Frédéric A. Dreyer, Lina Necib, Gregory Soyez, and Jesse Thaler, *Recursive Soft Drop*, JHEP 1806:093 (2018) [arXiv:1804.03657].
- [88] * Eric M. Metodiev and Jesse Thaler, *On the Topic of Jets: Disentangling Quarks and Gluons at Colliders*, Phys. Rev. Lett. 120:241602 (2018) [arXiv:1802.00008].
- [87] * Patrick T. Komiske, Eric M. Metodiev, and Jesse Thaler, *Energy Flow Polynomials: A Complete Linear Basis for Jet Substructure*, JHEP 1804:013 (2018) [arXiv:1712.07124].
- [86] Evan Coleman, Marat Freytsis, Andreas Hinzmann, Meenakshi Narain, Jesse Thaler, Nhan Tran, and Caterina Vernieri, *The Importance of Calorimetry for Highly-Boosted Jet Substructure*, JINST 13:T01003 (2018) [arXiv:1709.08705].
- [85] * Eric M. Metodiev, Benjamin Nachman, and Jesse Thaler, *Classification Without Labels: Learning from Mixed Samples in High Energy Physics*, JHEP 1710:174 (2017) [arXiv:1708.02949].

- [84] [†] Christopher Frye, Andrew J. Larkoski, Jesse Thaler, and Kevin Zhou, *Casimir Meets Poisson: Improved Quark/Gluon Discrimination with Counting Observables*, JHEP 1709:085 (2017) [arXiv:1704.06266].
- [83] ^{†§} Aashish Tripathy, Wei Xue, Andrew Larkoski, Simone Marzani, and Jesse Thaler, *Jet Substructure Studies with CMS Open Data*, Phys. Rev. D96:074003 (2017) [arXiv:1704.05842].
- [82] ^{*†} Benjamin T. Elder, Massimiliano Procura, Jesse Thaler, Wouter J. Waalewijn, and Kevin Zhou, *Generalized Fragmentation Functions for Fractal Jet Observables*, JHEP 1706:085 (2017) [arXiv:1704.05456].
- [81] [†] Andrew Larkoski, Simone Marzani, Jesse Thaler, Aashish Tripathy, and Wei Xue, *Exposing the QCD Splitting Function with CMS Open Data*, Phys. Rev. Lett. 119:132003 (2017) [arXiv:1704.05066].
- [80] Philippe Gras, Stefan Höche, Deepak Kar, Andrew Larkoski, Leif Lönnblad, Simon Plätzer, Andrzej Siódmiok, Peter Skands, Gregory Soyez, and Jesse Thaler, *Systematics of Quark/Gluon Tagging*, JHEP 1707:091 (2017) [arXiv:1704.03878].
- [79] Yevgeny Kats, Matthew McCullough, Gilad Perez, Yotam Soreq, and Jesse Thaler, *Colorful Twisted Top Partners and Partnerium at the LHC*, JHEP 1706:126 (2017) [arXiv:1704.03393].
- [78] Philip Ilten, Nicholas L. Rodd, Jesse Thaler, and Mike Williams, *Disentangling Heavy Flavor at Colliders*, Phys. Rev. D96:054019 (2017) [arXiv:1702.02947].
- [77] ^{*} Ian Moulton, Lina Necib, and Jesse Thaler, *New Angles on Energy Correlation Functions*, JHEP 1612:153 (2016) [arXiv:1609.07483].
- [76] Fabio Maltoni, Michele Selvaggi, and Jesse Thaler, *Resurrecting the Dead Cone*, Phys. Rev. D94:054015 (2016) [arXiv:1606.03449].
- [75] Philip Ilten, Yotam Soreq, Jesse Thaler, Mike Williams, and Wei Xue, *Inclusive Dark Photon Search at LHCb*, Phys. Rev. Lett. 116:251803 (2016) [arXiv:1603.08926].
- [74] Yonatan Kahn, Benjamin R. Safdi, and Jesse Thaler, *Broadband and Resonant Approaches to Axion Dark Matter Detection*, Phys. Rev. Lett. 117:141801 (2016) [arXiv:1602.01086].
- [73] Sergio Ferrara, Renata Kallosh, and Jesse Thaler, *Cosmology with Orthogonal Nilpotent Superfields*, Phys. Rev. D93:043516 (2016) [arXiv:1512.00545].
- [72] Philip Ilten, Jesse Thaler, Mike Williams, and Wei Xue, *Dark Photons from Charm Mesons at LHCb*, Phys. Rev. D92:115017 (2015) [arXiv:1509.06765].
- [71] [†] Jesse Thaler and Thomas F. Wilkason, *Resolving Boosted Jets with XCone*, JHEP 1512:051 (2015) [arXiv:1508.01518].
- [70] [†] Iain W. Stewart, Frank J. Tackmann, Jesse Thaler, Christopher K. Vermilion, and Thomas F. Wilkason, *XCone: N-jettiness as an Exclusive Cone Jet Algorithm*, JHEP 1511:072 (2015) [arXiv:1508.01516].
- [69] ^{*} Nayara Fonseca, Lina Necib, and Jesse Thaler, *Dark Matter, Shared Asymmetries, and Galactic Gamma Ray Signals*, JCAP 1602:052 (2016) [arXiv:1507.08295].

- [68] Jesse Thaler, *Separated at Birth: Jet Maximization, Axis Minimization, and Stable Cone Finding*, Phys. Rev. D92:074001 (2015) [arXiv:1506.07876].
- [67] * Yonatan Kahn, Daniel A. Roberts, and Jesse Thaler, *The Goldstone and Goldstino of Supersymmetric Inflation*, JHEP 1510:001 (2015) [arXiv:1504.05958].
- [66] Andrew J. Larkoski, Simone Marzani, and Jesse Thaler, *Sudakov Safety in Perturbative QCD*, Phys. Rev. D91:111501 (2015) [arXiv:1502.01719].
- [65] Daniele Bertolini, Jesse Thaler, and Jonathan R. Walsh, *The First Calculation of Fractional Jets*, JHEP 1505:008 (2015) [arXiv:1501.01965].
- [64] * Yonatan Kahn, Gordan Krnjaic, Jesse Thaler, and Matthew Toups, *DAEdALUS and Dark Matter Detection*, Phys. Rev. D91:055006 (2015) [arXiv:1411.1055].
- [63] Andrew J. Larkoski, Jesse Thaler, and Wouter J. Waalewijn, *Gaining (Mutual) Information about Quark/Gluon Discrimination*, JHEP 1411:129 (2014) [arXiv:1408.3122].
- [62] Andrew J. Larkoski and Jesse Thaler, *Aspects of Jets at 100 TeV*, Phys. Rev. D90:034010 (2014) [arXiv:1406.7011].
- [61] * Kaustubh Agashe, Yanou Cui, Lina Necib, and Jesse Thaler, *(In)direct Detection of Boosted Dark Matter*, JCAP 1410:062 (2014) [arXiv:1405.7370].
- [60] Andrew J. Larkoski, Simone Marzani, Gregory Soyez, and Jesse Thaler, *Soft Drop*, JHEP 1405:146 (2014) [arXiv:1402.2657].
- [59] Andrew J. Larkoski, Duff Neill, and Jesse Thaler, *Jet Shapes with the Broadening Axis*, JHEP 1404:017 (2014) [arXiv:1401.2158].
- [58] * Daniele Bertolini, Tucker Chan, and Jesse Thaler, *Jet Observables Without Jet Algorithms*, JHEP 1404:013 (2014) [arXiv:1310.7584].
- [57] * Yonatan Kahn, Matthew McCullough, and Jesse Thaler, *Auxiliary Gauge Mediation: A New Route to Mini-Split Supersymmetry*, JHEP 1311:161 (2013) [arXiv:1308.3490].
- [56] * Francesco D'Eramo, Jesse Thaler, and Zoe Thomas, *Anomaly Mediation from Unbroken Supergravity*, JHEP 1309:125 (2013) [arXiv:1307.3251].
- [55] Andrew J. Larkoski and Jesse Thaler, *Unsafe but Calculable: Ratios of Angularities in Perturbative QCD*, JHEP 1309:137 (2013) [arXiv:1307.1699].
- [54] Hsi-Ming Chang, Massimiliano Procura, Jesse Thaler, and Wouter J. Waalewijn, *Calculating Track Thrust with Track Functions*, Phys. Rev. D88:034030 (2013) [arXiv:1306.6630].
- [53] John Kearney, Aaron Pierce, and Jesse Thaler, *Exotic Top Partners and Little Higgs*, JHEP 1310:230 (2013) [arXiv:1306.4314].
- [52] Andrew J. Larkoski, Gavin P. Salam, and Jesse Thaler, *Energy Correlation Functions for Jet Substructure*, JHEP 1306:108 (2013) [arXiv:1305.0007].
- [51] John Kearney, Aaron Pierce, and Jesse Thaler, *Top Partner Probes of Extended Higgs Sectors*, JHEP 1308:130 (2013) [arXiv:1304.4233].

- [50] Hsi-Ming Chang, Massimiliano Procura, Jesse Thaler, and Wouter J. Waalewijn, *Calculating Track-Based Observables for the LHC*, Phys. Rev. Lett. 111:102002 (2013) [arXiv:1303.6637].
- [49] * Daniele Bertolini, Jesse Thaler, and Zoe Thomas, *Super-Tricks for Superspace*, TASI 2012 [arXiv:1302.6229].
- [48] * Francesco D’Eramo, Matthew McCullough, and Jesse Thaler, *Multiple Gamma Lines from Semi-Annihilation*, JCAP 1304:030 (2013) [arXiv:1210.7817].
- [47] Vicent Mateu, Iain W. Stewart, and Jesse Thaler, *Power Corrections to Event Shapes with Mass-Dependent Operators*, Phys. Rev. D87:014025 (2013) [arXiv:1209.3781].
- [46] * Yonatan Kahn and Jesse Thaler, *Searching for an Invisible A' Vector Boson with DarkLight*, Phys. Rev. D86:115012 (2012) [arXiv:1209.0777].
- [45] Ilya Feige, Matthew D. Schwartz, Iain W. Stewart, and Jesse Thaler, *Precision Jet Substructure from Boosted Event Shapes*, Phys. Rev. Lett. 109:092001 (2012) [arXiv:1204.3898].
- [44] Nathaniel Craig, Matthew McCullough, and Jesse Thaler, *Flavor Mediation Delivers Natural SUSY*, JHEP 1206:046 (2012) [arXiv:1203.1622].
- [43] * Yonatan Kahn and Jesse Thaler, *Locality in Theory Space*, JHEP 1207:007 (2012) [arXiv:1202.5491].
- [42] * Francesco D’Eramo, Jesse Thaler, and Zoe Thomas, *The Two Faces of Anomaly Mediation*, JHEP 1206:151 (2012) [arXiv:1202.1280].
- [41] Nathaniel Craig, Matthew McCullough, and Jesse Thaler, *The New Flavor of Higgsed Gauge Mediation*, JHEP 1203:049 (2012) [arXiv:1201.2179].
- [40] *[†] Francesco D’Eramo, Lin Fei, and Jesse Thaler, *Dark Matter Assimilation into the Baryon Asymmetry*, JCAP 1203:010 (2012) [arXiv:1111.5615].
- [39] * Daniele Bertolini, Keith Rehermann, and Jesse Thaler, *Visible Supersymmetry Breaking and an Invisible Higgs*, JHEP 1204:130 (2012) [arXiv:1111.0628].
- [38] [†] Jesse Thaler and Ken Van Tilburg, *Maximizing Boosted Top Identification by Minimizing N -subjettiness*, JHEP 1202:093 (2012) [arXiv:1108.2701].
- [37] Nathaniel Craig, Daniel Stolarski, and Jesse Thaler, *A Fat Higgs with a Magnetic Personality*, JHEP 1111:145 (2011) [arXiv:1106.2164].
- [36] * Clifford Cheung, Francesco D’Eramo, and Jesse Thaler, *The Spectrum of Goldstini and Modulini*, JHEP 1108:115 (2011) [arXiv:1104.2600].
- [35] * Clifford Cheung, Francesco D’Eramo, and Jesse Thaler, *Supergravity Computations without Gravity Complications*, Phys. Rev. D84:085012 (2011) [arXiv:1104.2598].
- [34] * Jesse Thaler and Zoe Thomas, *Goldstini Can Give the Higgs a Boost*, JHEP 1107:060 (2011) [arXiv:1103.1631].
- [33] * Jesse Thaler and Ken Van Tilburg, *Identifying Boosted Objects with N -subjettiness*, JHEP 1103:015 (2011) [arXiv:1011.2268].

- [32] Martin Schmaltz, Daniel Stolarski, and Jesse Thaler, *The Bestest Little Higgs*, JHEP 1009:018 (2010) [arXiv:1006.1356].
- [31] Clifford Cheung, Jeremy Mardon, Yasunori Nomura, and Jesse Thaler, *A Definitive Signal of Multiple Supersymmetry Breaking*, JHEP 1007:035 (2010) [arXiv:1004.4637].
- [30] JiJi Fan, Jesse Thaler, and Lian-Tao Wang, *Dark Matter from Dynamical SUSY Breaking*, JHEP 1006:045 (2010) [arXiv:1004.0008].
- [29] * Francesco D'Eramo and Jesse Thaler, *Semi-annihilation of Dark Matter*, JHEP 1006:109 (2010) [arXiv:1003.5912].
- [28] Clifford Cheung, Yasunori Nomura, and Jesse Thaler, *Goldstini*, JHEP 1003:073 (2010) [arXiv:1002.1967].
- [27] David Krohn, Jesse Thaler, and Lian-Tao Wang, *Jet Trimming*, JHEP 1002:084 (2010) [arXiv:0912.1342].
- [26] Marat Freytsis, Zoltan Ligeti, and Jesse Thaler, *Constraining the Axion Portal with $B \rightarrow K l^+ l^-$* , Phys. Rev. D81:034001 (2010) [arXiv:0911.5355].
- [25] Christian W. Bauer, Zoltan Ligeti, Martin Schmaltz, Jesse Thaler, and Devin G.E. Walker, *Supermodels for early LHC*, Phys. Lett. B 690:280-288 (2010) [arXiv:0909.5213].
- [24] Marat Freytsis, Grigory Ovanessian, and Jesse Thaler, *Dark Force Detection in Low Energy e - p Collisions*, JHEP 1001:111 (2010) [arXiv:0909.2862].
- [23] Jeremy Mardon, Yasunori Nomura, and Jesse Thaler, *Cosmic Signals from the Hidden Sector*, Phys. Rev. D80:035013 (2009) [arXiv:0905.3749].
- [22] David Krohn, Jesse Thaler, and Lian-Tao Wang, *Jets with Variable R* , JHEP 0906:059 (2009) [arXiv:0903.0392].
- [21] Jeremy Mardon, Yasunori Nomura, Daniel Stolarski, and Jesse Thaler, *Dark Matter Signals from Cascade Annihilations*, JCAP 0905:016 (2009) [arXiv:0901.2926].
- [20] Martin Schmaltz and Jesse Thaler, *Collective Quartics and Dangerous Singlets in Little Higgs*, JHEP 0903:137 (2009) [arXiv:0812.2477].
- [19] Yasunori Nomura and Jesse Thaler, *Dark Matter through the Axion Portal*, Phys. Rev. D79:075008 (2009) [arXiv:0810.5397].
- [18] David Poland and Jesse Thaler, *The Dark Top*, JHEP 0811:083 (2008) [arXiv:0808.1290].
- [17] Jesse Thaler and Lian-Tao Wang, *Strategies to Identify Boosted Tops*, JHEP 0807:092 (2008) [arXiv:0806.0023].
- [16] Christian W. Bauer, Frank J. Tackmann, and Jesse Thaler, *GenEvA (II): A phase space generator from a reweighted parton shower*, JHEP 0812:011 (2008) [arXiv:0801.4028].
- [15] Christian W. Bauer, Frank J. Tackmann, and Jesse Thaler, *GenEvA (I): A new framework for event generation*, JHEP 0812:010 (2008) [arXiv:0801.4026].
- [14] Yuval Grossman, Yosef Nir, Jesse Thaler, Tomer Volansky, and Jure Zupan, *Probing Minimal Flavor Violation at the LHC*, Phys. Rev. D76:096006 (2007) [arXiv:0706.1845].

- [13] Nima Arkani-Hamed, Bruce Knuteson, Stephen Mrenna, Philip Schuster, Jesse Thaler, Natalia Toro, and Lian-Tao Wang, *MARMOSET: The Path from LHC Data to the New Standard Model via On-Shell Effective Theories* [arXiv:hep-ph/0703088].
- [12] Aaron Pierce and Jesse Thaler, *Natural Dark Matter from an Unnatural Higgs Boson and New Colored Particles at the TeV Scale*, JHEP 0708:026 (2007) [arXiv:hep-ph/0703056].
- [11] Aaron Pierce, Jesse Thaler, and Lian-Tao Wang, *Disentangling Dimension Six Operators through Di-Higgs Boson Production*, JHEP 0705:070 (2007) [arXiv:hep-ph/0609049].
- [10] Hsin-Chia Cheng, Jesse Thaler, and Lian-Tao Wang, *Little M-theory*, JHEP 0609:003 (2006) [arXiv:hep-ph/0607205].
- [9] Clifford Cheung and Jesse Thaler, *(Reverse) Engineering Vacuum Alignment*, JHEP 0608:016 (2006) [arXiv:hep-ph/0604259].
- [8] Aaron Pierce and Jesse Thaler, *Prospects for Mirage Mediation*, JHEP 0609:017 (2006) [arXiv:hep-ph/0604192].
- [7] Hsin-Chia Cheng, Markus A. Luty, Shinji Mukohyama, and Jesse Thaler, *Spontaneous Lorentz Breaking at High Energies*, JHEP 0605:076 (2006) [arXiv:hep-th/0603010].
- [6] Nima Arkani-Hamed, Gordon L. Kane, Jesse Thaler, and Lian-Tao Wang, *Supersymmetry and the LHC Inverse Problem*, JHEP 0608:070 (2006) [arXiv:hep-ph/0512190].
- [5] Yuval Grossman, Can Kilic, Jesse Thaler, and Devin G. E. Walker, *Neutrino Constraints on Spontaneous Lorentz Violation*, Phys. Rev. D72:125001 (2005) [arXiv:hep-ph/0506216].
- [4] Jesse Thaler, *Little Technicolor*, JHEP 0507:024 (2005) [arXiv:hep-ph/0502175].
- [3] Jesse Thaler and Itay Yavin, *The Littlest Higgs in Anti-de Sitter Space*, JHEP 0508:022 (2005) [arXiv:hep-ph/0501036].
- [2] Nima Arkani-Hamed, Hsin-Chia Cheng, Markus A. Luty, and Jesse Thaler, *Universal Dynamics of Spontaneous Lorentz Violation and a New Spin-Dependent Inverse-Square Law Force*, JHEP 0507:029 (2005) [arXiv:hep-ph/0407034].
- [1] Antal Jevicki and Jesse Thaler, *Dynamics of black hole formation in an exactly solvable model*, Phys. Rev. D66 024041 (2002) [arXiv:hep-th/0203172].